Lesson notes · for you

Exploring a dataset

Data science · Level 2 · Lesson 4

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Follow a five-question routine.
- Spot gaps and oddities.
- Write findings with evidence.

Activity: Five questions, three findings

1. Warm-up (5 min): Look at a big table and say the first thing you would check.
2. Explore (10 min): With a partner, work through the five questions for a small table.
3. Write (10 min): Write three findings with numbers and sources, and one next question.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Which question found a surprise?
- What makes a finding good?
- What is your next question?

Check yourself

- Write one finding from today with a number and a source.

Try it this week
Pick a small table, such as a week of weather or your class's library loans, and go through the five questions. Write three findings with numbers and one next question.

Worksheet

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Name _____ Class _____ Date _____

1 **Which is the better finding?**

◯ Attendance is bad ◯ Attendance changes during the year

◯ Attendance falls to a median of 80% in July

2 **Put the five questions in order.**

☐ Centre and spread

☐ What is in it?

☐ What shape?

☐ Relationships

☐ Gaps and oddities

3 **Write summary(xs) to return the smallest value, the median and the largest. xs has an odd length.**

```
def summary(xs):
    return 0, 0, 0

print(summary([7, 9, 6, 25, 8]))
```

Write your program:

4 **Your screen-time column is skewed right. What should you report?**

◯ Only the largest value ◯ Only the mean ◯ The median and the range

5 **Which are good findings? Choose all that apply.**

☐ Two days have no screen time recorded. ☐ Sleep is important.

☐ On school nights my median sleep is 7 hours. ☐ I'm always tired.

✎ **What would you like to explore about your own week, if you kept the data?**

Check your answers

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1 **Attendance falls to a median of 80% in July**

A good finding has a number, a time and a source.

2 **Top to bottom: 3, 1, 2, 5, 4**

Get to know the table, then each column, then problems, then comparisons.

3

```
def summary(xs):
    s = sorted(xs)
    return s[0], s[len(s) // 2], s[-1]

print(summary([7, 9, 6, 25, 8]))
```

After sorting, the first, middle and last items are the minimum, median and maximum.

4 **The median and the range**

A skewed column needs a middle that resists the tail, plus a spread.

5 **On school nights my median sleep is 7 hours.; Two days have no screen time recorded.**

Good findings are specific and checkable.